

Visión por Computadora Inspección de Apilamiento de Pañales.

EDA & Fase de Preparación y Procesamiento de Datos



Online
Universidad
Espíritu Santo

**Inteligencia
Artificial**

Presentación 1

Nombre de la materia

PROYECTO INTEGRADOR EN IA – MIAR0545

Nombre del alumno

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Fecha

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Introducción

Temas

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Proyecto

Tema 2

Estadística Descriptiva y
EDA

Tema 3

Preparación y
Procesamiento

Tema 4

Conclusiones y Caso de
Estudio

Repositorio:

<https://github.com/DAVOALEJO1987/VISION-ARTIFICIAL-DE-APILAMIENTO-DE-PANALES-EDA-Fase-de-Preparaci-n-y-Procesamiento-de-Datos.git>

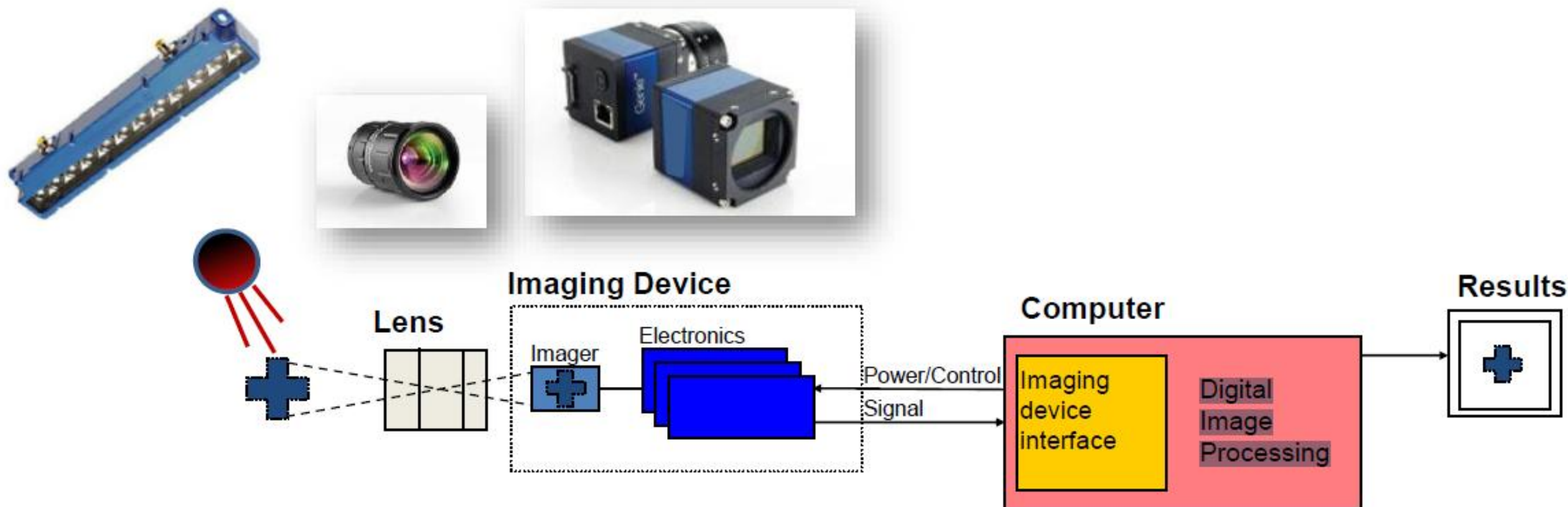
Referencias:

- [1] Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems (3rd ed.). O'Reilly Media.
- [2] Machine-Vision-Systems-Design A3 ACADEMY

Generalidades

01

Visión por computador, 6w, desarrollar un prototipo funcional de visión con CNN y cámara industrial que, tras normalización, filtrado, aumento y segmentación, verifique automáticamente el apilamiento de pañales antes del empaque secundario, logrando $mAP \geq 0.70$, $IoU > 0.50$ y > 30 FPS



Tema 1:

VISION ARTIFICIAL DE APILAMINTO DE PANALES

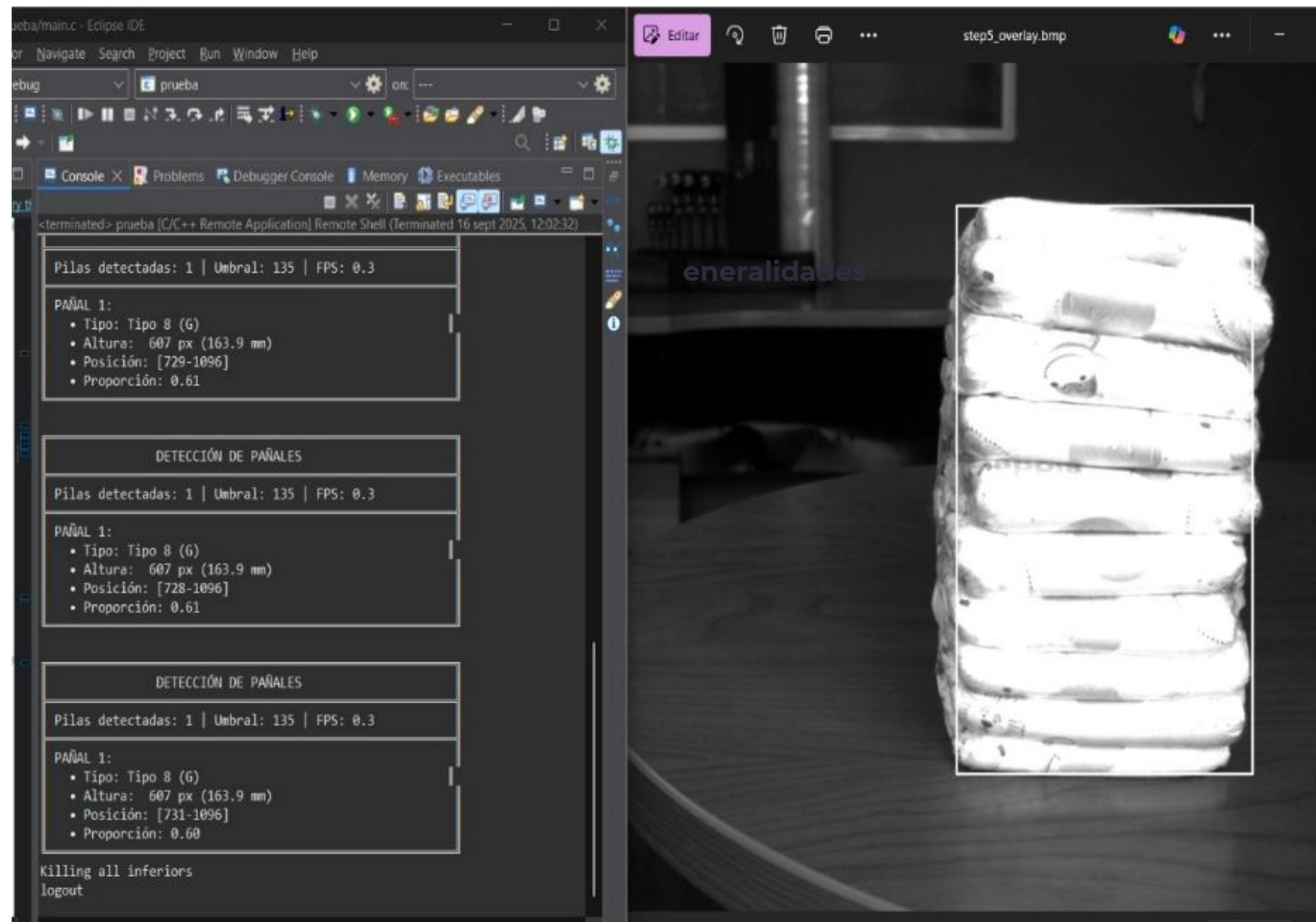


Diagrama de Trabajo VC

Tema 2:

Definición y Selección

Analisis

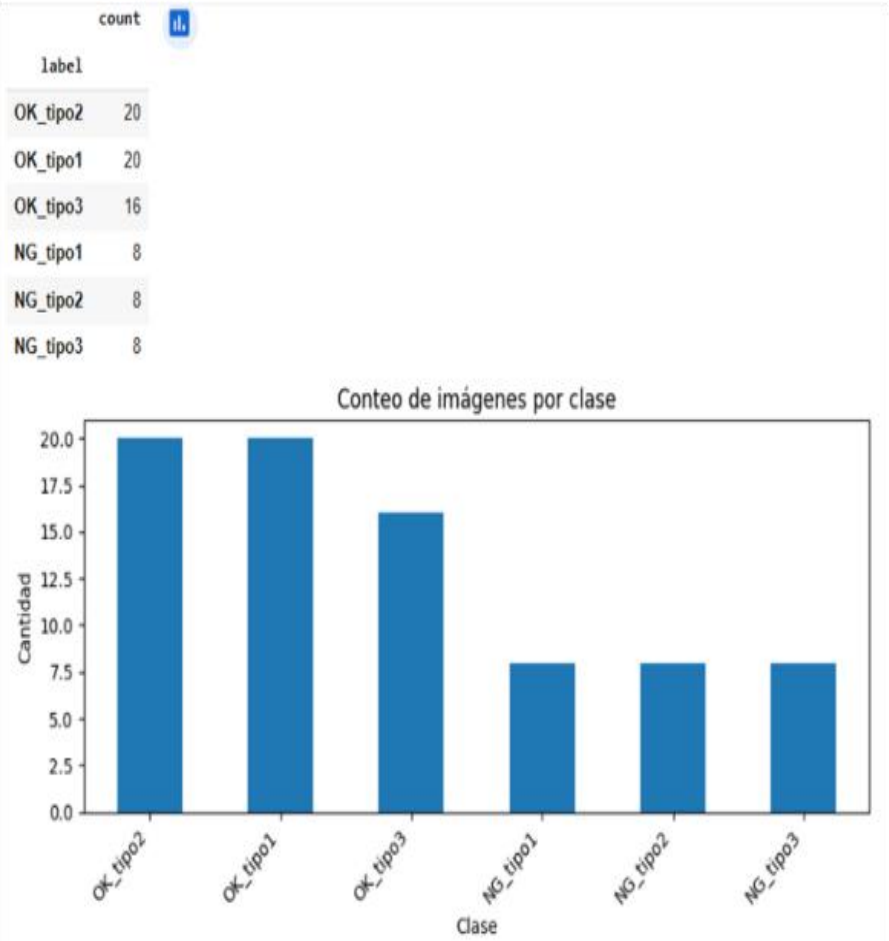
EDA

Repositorio:
<https://github.com/DAVOALEJO1987/VISION-ARTIFICIAL-DE-APILAMIENTO-DE-PANALES-EDA-Fase-de-Preparaci-n-y-Procesamiento-de-Datos.git>

=== Exploración inicial ===
Total de imágenes: 80

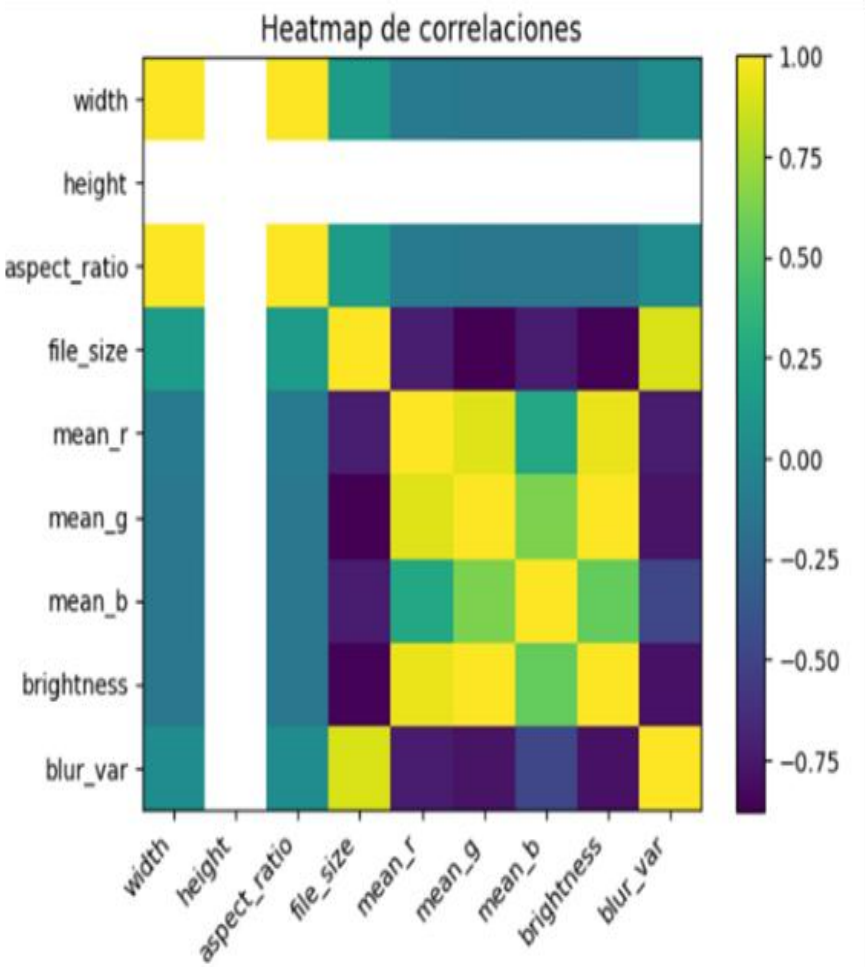
	filepath	label	width	height	aspect_ratio	file_size	mean_r	mean_g	mean_b	brightness	blur_var	phash
0	/content/drive/MyDrive/Proyecto Final Maestría...	NG_tipo1	640	640	1.0	58761	175.354778	165.320083	155.790190	166.765401	32.886587	None
1	/content/drive/MyDrive/Proyecto Final Maestría...	NG_tipo1	640	640	1.0	61672	176.710542	166.370843	156.724167	167.879995	34.523157	None
2	/content/drive/MyDrive/Proyecto Final Maestría...	NG_tipo1	640	640	1.0	59847	177.415769	166.435310	156.969580	168.086330	32.723040	None
3	/content/drive/MyDrive/Proyecto Final Maestría...	NG_tipo1	640	640	1.0	61135	173.885386	163.182036	154.662358	164.842448	46.404414	None
4	/content/drive/MyDrive/Proyecto Final Maestría...	NG_tipo1	640	640	1.0	60574	173.334446	163.868684	154.643420	165.215041	35.123552	None

	filepath	label	width	height	aspect_ratio	file_size	mean_r	mean_g	mean_b	brightness	blur_var	phash
count	80	80	80.000000	80.0	80.000000	80.000000	80.000000	80.000000	80.000000	80.000000	80.000000	0
unique	80	6	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	0
top	/content/drive/MyDrive/Proyecto Final Maestría...	OK_tipo2	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
freq	1	20	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
mean	NaN	NaN	638.000000	640.0	0.996875	86449.075000	155.225392	149.085875	143.070747	149.956844	58.315561	NaN
std	NaN	NaN	17.880544	0.0	0.027951	33979.764564	27.710634	16.973453	14.667806	18.250129	40.139661	NaN
min	NaN	NaN	480.000000	640.0	0.750000	41512.000000	73.056628	106.734712	117.706089	101.971209	11.404858	NaN
25%	NaN	NaN	640.000000	640.0	1.000000	58791.000000	149.801732	136.046843	126.678857	137.564281	31.041429	NaN
50%	NaN	NaN	640.000000	640.0	1.000000	62497.000000	167.596895	157.500791	150.324977	159.260867	40.736577	NaN
75%	NaN	NaN	640.000000	640.0	1.000000	124930.750000	172.230848	163.512297	154.993022	164.810960	93.188066	NaN
max	NaN	NaN	640.000000	640.0	1.000000	142906.000000	179.150905	168.281582	160.398887	170.623269	157.862779	NaN



=== Matriz de correlación ===

	width	height	aspect_ratio	file_size	mean_r	mean_g	mean_b	brightness	blur_var
width	1.000000	NaN	1.000000	0.149728	-0.097725	-0.128042	-0.133753	-0.124486	0.035331
height	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
aspect_ratio	1.000000	NaN	1.000000	0.149728	-0.097725	-0.128042	-0.133753	-0.124486	0.035331
file_size	0.149728	NaN	0.149728	1.000000	-0.719998	-0.876641	-0.735850	-0.858303	0.896849
mean_r	-0.097725	NaN	-0.097725	-0.719998	1.000000	0.912014	0.258419	0.944540	-0.731635
mean_g	-0.128042	NaN	-0.128042	-0.876641	0.912014	1.000000	0.627783	0.996088	-0.778995
mean_b	-0.133753	NaN	-0.133753	-0.735850	0.258419	0.627783	1.000000	0.559054	-0.479143
brightness	-0.124486	NaN	-0.124486	-0.858303	0.944540	0.996088	0.559054	1.000000	-0.782213
blur_var	0.035331	NaN	0.035331	0.896849	-0.731635	-0.778995	-0.479143	-0.782213	1.000000

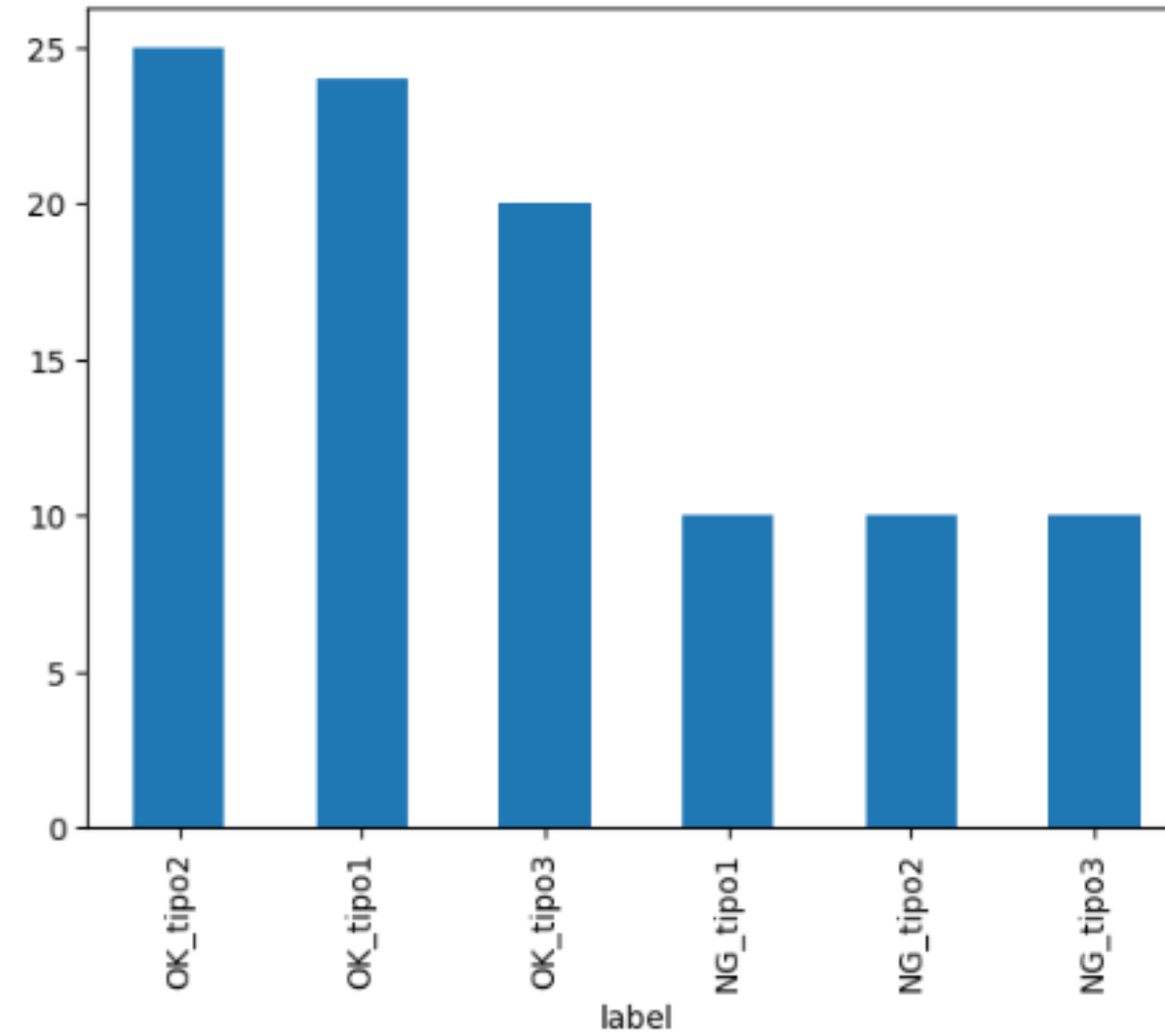


Tema 3:

Pipeline de Limpieza de Datos &

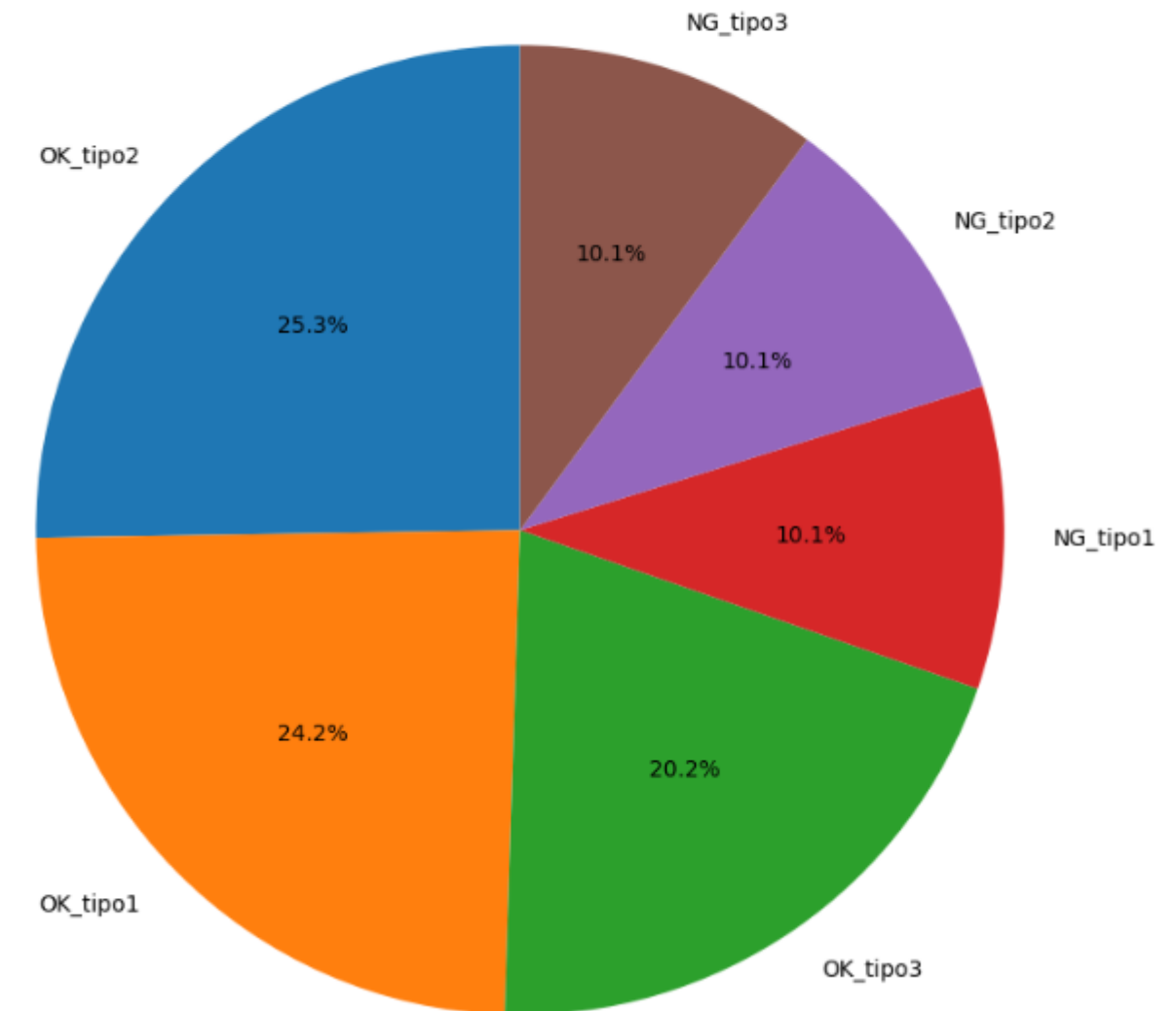
Balanceo

Total imágenes: 99



	filepath	label	size
0	data/NG_tipo1/tipo1_NG_0003.jpg	NG_tipo1	(640, 640)
1	data/NG_tipo1/tipo1_NG_0010.jpg	NG_tipo1	(640, 640)
2	data/NG_tipo1/tipo1_NG_0006.jpg	NG_tipo1	(640, 640)
3	data/NG_tipo1/tipo1_NG_0005.jpg	NG_tipo1	(640, 640)
4	data/NG_tipo1/tipo1_NG_0004.jpg	NG_tipo1	(640, 640)

Distribución de Clases

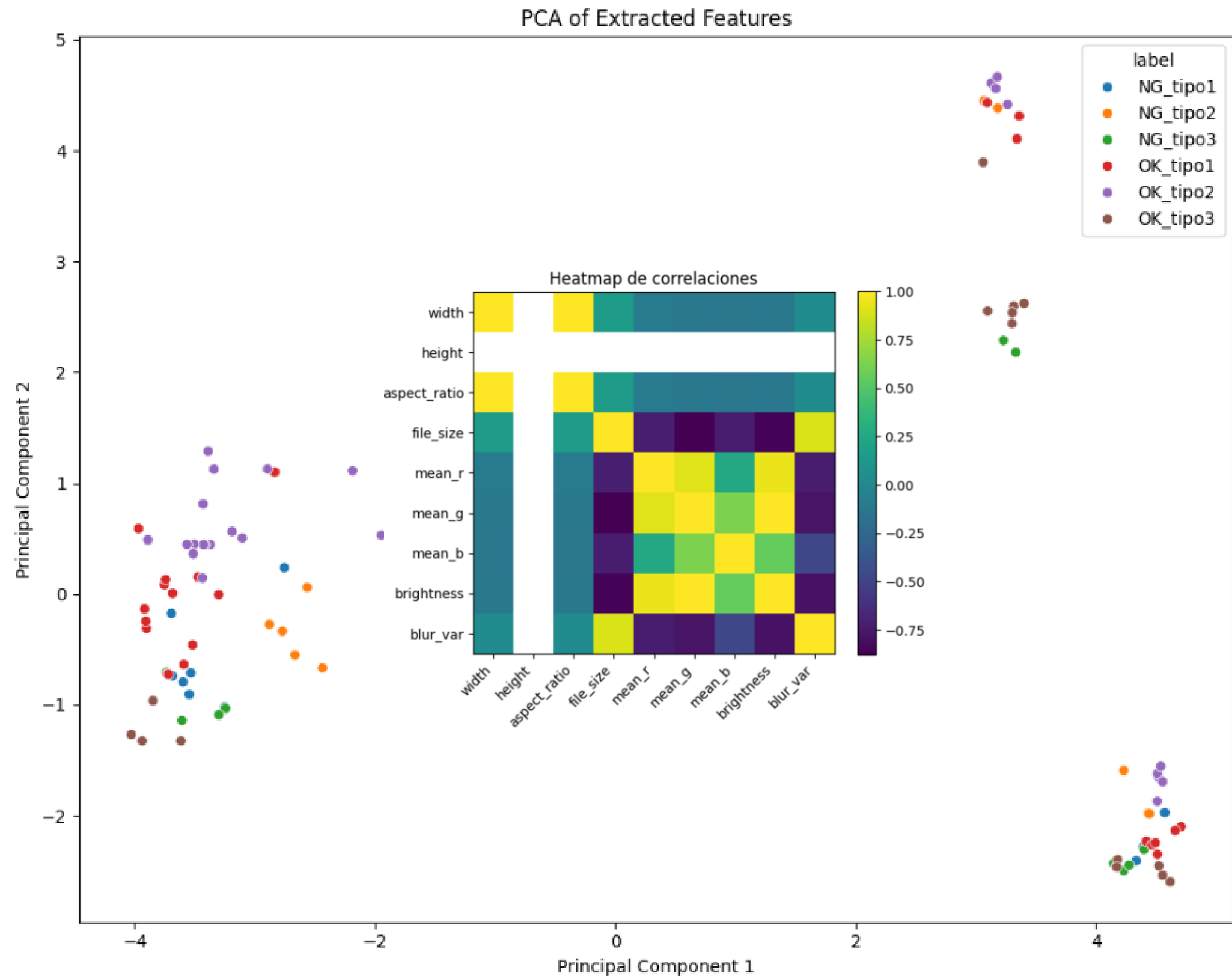


Tema 4:

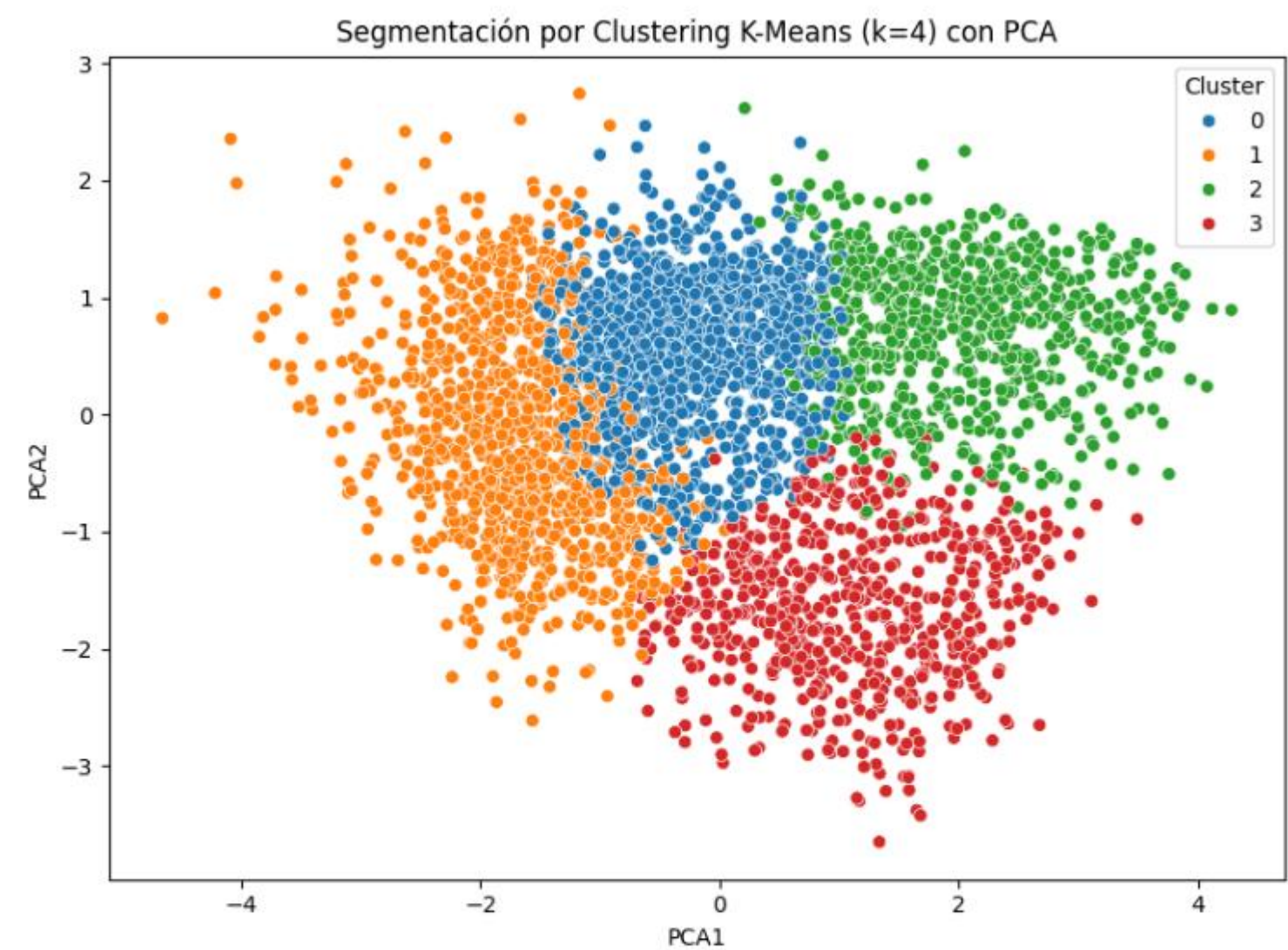
Conclusiones

```
clf = RandomForestClassifier()  
clf.fit(X_train, y_train)  
preds = clf.predict(X_test)  
print(classification_report(y_test, preds))
```

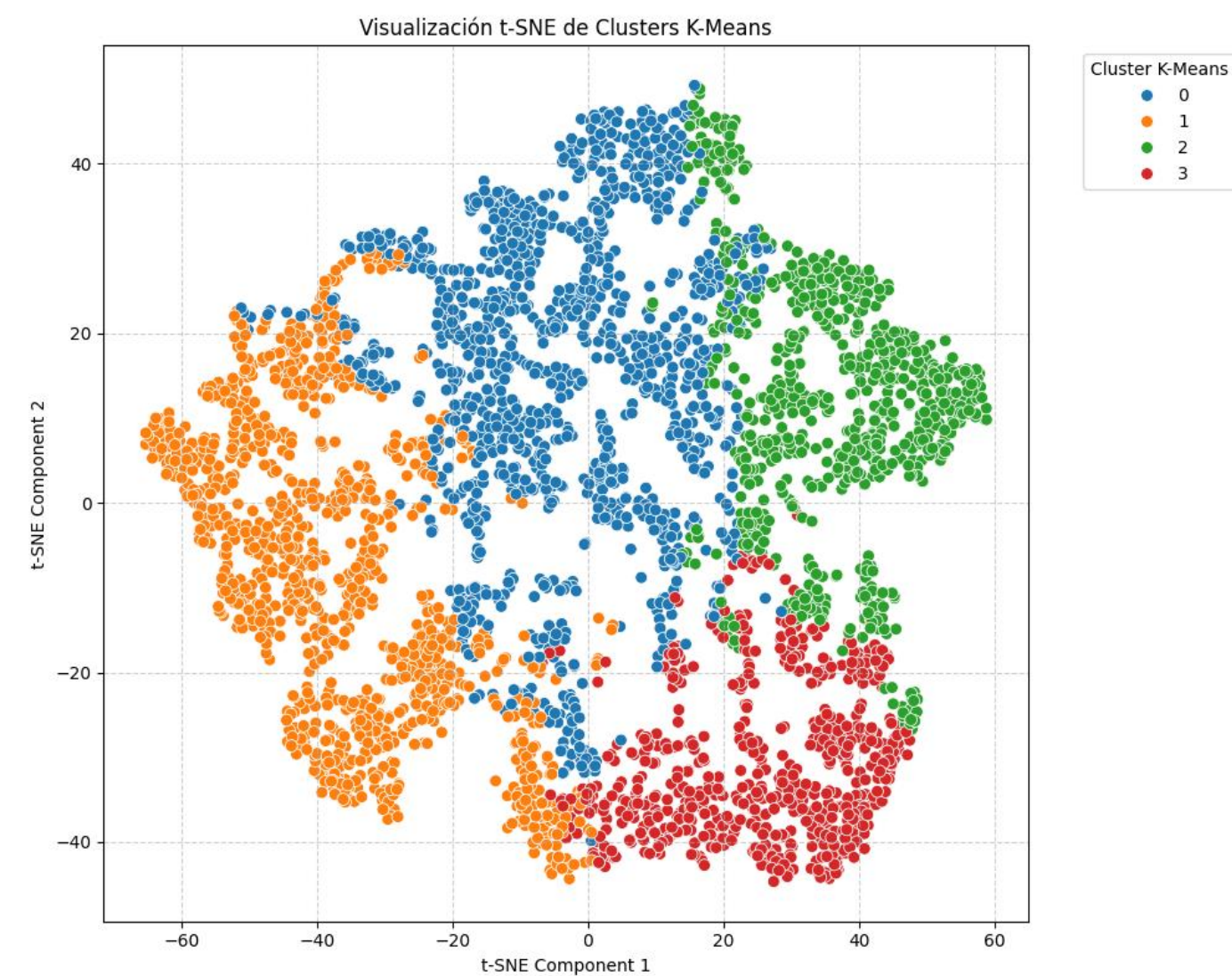
	precision	recall	f1-score	support
NG_tipo1	1.00	0.50	0.67	2
NG_tipo2	1.00	0.50	0.67	2
NG_tipo3	1.00	0.50	0.67	2
OK_tipo1	0.83	1.00	0.91	5
OK_tipo2	0.83	1.00	0.91	5
OK_tipo3	0.80	1.00	0.89	4
accuracy			0.85	20
macro avg	0.91	0.75	0.78	20
weighted avg	0.88	0.85	0.83	20



Siguientes pasos



	global_radiation	precipitation	sunshine	temp_mean	humidity
KMeans_Cluster					
0	1.736848	0.400428	2.938765	-1.616955	0.946428
1	1.004649	1.237731	1.074446	-10.013469	0.948625
2	2.924340	0.152463	10.688435	1.329388	0.804585
3	1.356258	0.061726	8.517258	-8.165323	0.565726



	global_radiation	precipitation	sunshine	temp_mean	humidity
KMeans_Cluster					
0	1.736848	0.400428	2.938765	-1.616955	0.946428
1	1.004649	1.237731	1.074446	-10.013469	0.948625
2	2.924340	0.152463	10.688435	1.329388	0.804585
3	1.356258	0.061726	8.517258	-8.165323	0.565726

Probando el ML

K-Means (0.25) ofrece mejor definición de grupos y es más confiable para interpretar y clasificar condiciones climáticas.

